Charles Grassi

Graphics Programmer & Technical Artist

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Shaders, lighting and VFX for games, VR and live installations, made with the care of a painter and the rigour of a profiler.

EXPERIENCE

2025–now

Live visuals, VR and AI



Blumhouse Enhanced Cinema

HYPERDRYFT — Co-founder & Graphics Lead

A live-visuals engine on WebGPU, delivered for a global brand's event booth.

Nexus Studios — Technical Artist & Graphics Programmer

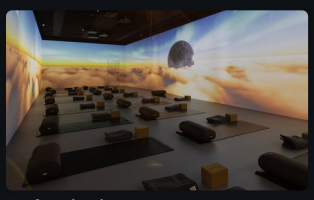
Blumhouse Enhanced Cinema for Meta Quest, nominated for a 2026 Primetime Emmy. Real-time global illumination for VR.

Events Arena — Chief Technology Officer

Built the company's platform from scratch in Python, FastAPI and PostgreSQL: booking intake, seat allocation, pricing and ticket logistics.

2023–2024

Leading teams



360° projection room

SABDA — Chief Technology Officer

Led the team behind a 360° projection room, with real-time 3D across several servers.

Little Red Dog Games — Lead Graphics Programmer

Led a team of technical artists building the game's VFX in Unity.

22cans — Senior Graphics Programmer

In Peter Molyneux's studio: rendering for a game on phones and PCs, from its material library to foliage and lighting.

2020–2022

Shaders for games and apps



The Sandbox

Scapin' — Graphics Programmer

Wrote the avatar shader library, a lighting library and custom render passes, tuned for tile-based mobile GPUs: half precision, low bandwidth, a cheap UI blur.

The Sandbox — Graphics Programmer

Worked on the game's stylized look: clouds, outlines and shafts of light.

Ubisoft — Technical Artist

Shaders and VFX for a title in production, and tools for the art team to work with them.

2016–2020

Starting out

Broxilia — Indie game developer

Taught myself computer science and fell for real-time graphics: wrote my first shaders and released a small game on Steam.

IXCYS C# Developer

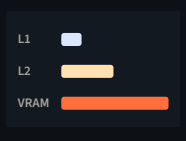
OTHER CLIENTS DREST · GN3RA · H3 Space · Lunchbox Entertainment

ARTICLES



Neural Shaders, Explained With Clouds

A small neural network, fitted to a path-traced reference, replaces the scattering approximation in a real-time cloud shader. **250,000+ views on Reddit and X.**



GPU Cache Hierarchy

Texture-fetch cost across the GPU memory hierarchy (L1, L2, VRAM), and how access patterns decide a shader's performance.

WHAT I USE

Rendering

Real-time GI · PBR · Ray marching · Compute shaders · GPU-driven rendering · Custom render passes · Post-processing · VFX · Neural shading · Mobile and VR optimisation

APIs and engines

WebGPU/WGSL · DX11/12 · Vulkan · Metal · OpenGL · HLSL · GLSL · Unreal (Niagara) · Unity (URP, HDRP)

Languages and tools

C++ · C# · Python · Rust · RenderDoc · NSight · PIX · Claude API · MCP · FastAPI · PostgreSQL